Giulio Calacoci

Principal Software Engineer · Tech Lead · AI Agent Infrastructure, PostgreSQL, Platform Engineering

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20 years in software, 15 in PostgreSQL · AI agent infrastructure since 2025 · built the CI/CD platform every EDB product ships through · Barman tech lead, team of up to 6

SUMMARY

Principal engineer, 20 years in software, 15 in PostgreSQL. Twice I built the continuous delivery platform every product shipped through: at 2ndQuadrant, then Foundation at EDB. I led Barman, PostgreSQL's open-source backup manager, with up to six engineers. Since 2024 I have built proofs of concept for the VP: a PostgreSQL optimizer, then AI agents from 2025, which led to pg-airman-mcp and EDB's AI team. Looking for a tech lead, principal or engineering manager role.

EXPERIENCE

EDB (EnterpriseDB) · Principal DevOps Architect → Principal Software Engineer

Sep 2020 – Aug 2026 · Florence, Italy (remote)

Joined through EDB's acquisition of 2ndQuadrant.

AI and agent infrastructure (Mar 2025 – Aug 2026)

- From March 2025, alongside the Barman lead: R&D and proofs of concept with the VP of the unit on AI agents and PostgreSQL. One of them became pg-airman-mcp, EDB's fork of the abandoned postgres-mcp project; I led it to 1.0 and production (see Open Source).
- AI team (Mar 2026 – Aug 2026): recruited into the team on the strength of pg-airman-mcp; part of the group that reviewed architecture decisions with VPs and SVPs.
- Co-designed the MCP connections of EDB AIDB; part of the team behind AIDB 7.7, which added an MCP endpoint and in-database agents to Postgres for EDB's enterprise customers.
- Designed session tracing and purpose logging, an audit trail linking each query to the agent session and its declared purpose; worked on agent governance with Postgres roles, grants and row-level security. Built Python components for EDB's internal Langflow fork.

Tech lead: Barman and the PostgreSQL optimization project (2023 – Mar 2026)

- Led the technical direction of Barman, the open-source backup and recovery manager for PostgreSQL (3,200+ GitHub stars), in Python: WAL archiving and streaming, base backups and point-in-time recovery, with backups to local disk or cloud storage (S3, Azure Blob, Google Cloud Storage). Team of up to six engineers; design docs (RFCs) and PR reviews; shipped about 23 releases (3.4 to 3.18), making the go/no-go call with the director.
- Built reliability into the product: integration with PostgreSQL failover, tested on Kubernetes clusters that simulate failovers and switchovers; incremental and resumable backups; designed and led immutable (WORM) backup storage support.
- Second-level (expert) support engineer for backup and recovery on customer installations, several with databases above 10 TB; post-mortems were routine.
- PostgreSQL optimization project (2024 – Mar 2026): began as a proof of concept I built for the VP of the unit; after a presentation to executive leadership it became a project, which I led, mentoring a junior developer. It runs in the control plane of EDB's internal multi-tenant managed Postgres service (DBaaS) on Kubernetes, deployed through Helm and Terraform.
- Designed the technical interview my teams used for hiring. Hired three engineers into the Barman team and mentored them; they now write most of Barman's code.
- First adopter of GitHub Copilot in the Barman team, on EDB's enterprise GitHub; presented the setup to the team.

Principal DevOps Architect: Foundation, continuous delivery (2020 – 2023)

- Lead architect and creator of **Foundation**, EDB's GitHub-based framework to build, test, sign and publish every EDB product: 7 OS families and 4 CPU architectures, on GitHub Actions, AWS and Cloudsmith. Took it from zero to adoption by every product team; release preparation went from days to about a day.

- One of the owners of Foundation's AWS account; managed its infrastructure as code with Terraform and Helm, including the AWS Lambda functions that checked and distributed packages. Release controls: review and signing before publishing, penetration-tested.
- Technical lead of the team that merged EDB's and 2ndQuadrant's CI/CD infrastructures, working with the SVP on infrastructure decisions.
- Two direct reports (2020–2023).

2ndQuadrant · Continuous Delivery Manager

Sep 2016 – Sep 2020 · Prato, Italy

- Designed, built and ran the company's continuous delivery platform: a Jenkins framework that standardised builds for every product on every supported Linux distribution, with automated package signing and publishing. Release time went from days to hours.

2ndQuadrant · Python Developer and PostgreSQL Consultant

Aug 2011 – Sep 2020 · Prato, Italy

- Barman developer from 2013, for a time the project's only developer and its top contributor.
- PostgreSQL support engineer (L1 and L2) on the on-call rotation, weekend shifts included: ran customer calls during outages or joined them as the expert.
- Disaster recovery and high availability consultant: crisis support, consultancy, training, and high-performance database design.

Earlier

- **Devise.IT s.r.l.**, Software Analyst (2009–2011): web applications in Java, PHP and JavaScript. **grafica-web**, Chief Web Application Developer (2008–2009): led a small team.

OPEN SOURCE

- **Barman**: developer since 2013 (226 commits), tech lead 2023–2026.
- **pg-airman-mcp**: MCP server that gives AI agents governed access to PostgreSQL. Led EDB's fork to 1.0 and beyond (77 of 90 commits since the fork; built with AI-assisted development): OAuth 2.0, session tracing and purpose logging, hardened containers, five releases. Featured in a Red Hat Developer series on OpenShift AI, co-authored by EDB's AI lead.

INDEPENDENT PROJECTS (2026)

- **Agent orchestrator:** Claude plans and leads cheaper and free models in a sandbox; coding tasks start from a failing test, the checks re-run and I review the result. Data classes decide which models may see which data; routes come from logged pass rates, and replay benches re-run real past features against the reference build.
- **Guardrails:** a pre-tool hook blocks secret reads, pushes and destructive commands, with tests and an audit log; a fact gate checks drafted claims against source documents.
- **Autonomous data pipelines:** nightly agent pipelines collect web data from about 140 sources (up to about 18,000 items a night), back off on rate limits, detect bot walls, and have agents triage the results.
- **session-council** (open source, MIT): rules and a judge skill for long-running Claude Code sessions that review each other's work. Blind first rounds, reasons labelled fact or guess, a checked fact outweighs a majority, and votes only recommend: I approve every decision. Used to restructure this CV.

WRITING AND TALKS

- **Blog** at gcalacoci.com, cross-posted on dev.to: the series *Searching for a job in the AI era*, written while the search runs, and a post on Claude Code sessions reviewing each other.

- **Talks** (details and slides):
  - *Barman, past, present, and future of a pioneer community project*, PGConf.EU 2025, Riga
  - *From 0 to 100: Business continuity with PostgreSQL*, and a two-part workshop on backup and disaster recovery of PostgreSQL, DataOps Barcelona 2019
  - *PostgreSQL and Python, a match made in heaven*, EuroPython 2017
  - *Step up and face the disaster*, Dalibo PG Session #8, 2016
  - In Italian: PyCon Italia 2015 to 2018, PgDay.IT 2017, 5432...MeetUs! 2016

SKILLS

- **PostgreSQL:** high availability, replication, failover, backup and point-in-time recovery, disaster recovery, performance tuning, roles, grants and RLS
- **Platform:** Kubernetes (hands-on; working towards the CKA), Terraform, Helm, GitHub Actions, Jenkins, CI/CD and release automation, Docker, Linux; AWS (EC2, S3, IAM, VPC, ECR, RDS, Lambda, CloudWatch, App Runner; wide hands-on use); monitoring with Nagios, Zabbix, Grafana, some Prometheus
- **AI engineering:** LLM agents in production, MCP servers and tool access, agent access control (OAuth 2.0, Postgres roles, grants and RLS), session tracing and purpose logging, Langflow; multi-agent orchestration, evals and replay benches, guardrails with human review, spec-driven development and TDD with agents; daily agentic work with Claude Code (skills, memory, subagents)
- **Languages:** Python (senior), SQL, Go (working knowledge)
- **Ways of working:** end-to-end ownership; starting from no clear spec and shaping it (2ndQuadrant's CD platform, proofs of concept for the VP); design and decision documents; regular discussions with product managers and other principal engineers on their projects; customer feedback loops; open-source maintainership; hiring and mentoring; remote and asynchronous teams, documentation for everything

EDUCATION AND LANGUAGES

- Computer Science, Università degli Studi di Firenze, 2002–2008 · left to work as a developer
- **Italian** (native) · **English** (full professional proficiency)